



Ultrasound Haptic Feedback

Recently, a team of researchers at University of Tokyo have created a system which can give you the sensation of touch in mid-air. The system consist of cameras which track the motion of your hand, and as you approach this virtual object in midair, a focussed ultrasound wave hits your hand to impart the sensation of touch. Well, for now, these object can only be soft, as the intensity of these ultrasound waves is low – to prevent them from destroying your precious eardrums. Currently, these waves can impart the feeling of only a plain solid surface. However, researchers say that they can be manipulated to give sensation of different curvatures. Should be just in time for GameFloor 3!



Redefined

the perfect headshot. The bullet left a perfect, Bollywood-style smoke trail, and pwned the enemy for a splatter of holographic blood all over the floor. We even felt the weight of the virtual gun through the elaborate haptic feedback system of the GameFloor. Moreover, the new directional sound system of GameFloor 3 will make sure that the clatter of guns does not wake up neighbours in the middle of the night. The latest version of the consoles even ensures that each player gets his individual audio spotlight. Soon, a few friends joined in the fun our holographically present mates from our US office teamed up for a game of Soccer 2021. The new version of the console allows up to ten of your friends to be transported virtually to the confines of your room. The new feedback system makes the ball feel as good as real on your feet. But the best thing about the new GameFloor – if you are tired, pick up the controller and enjoy the game from the sofa without ever needing to pause; your place in the game is now taken by your customisable virtual avatar.

The excellent display and CIA-esque tracking takes the GameFloor to a whole new level. All this makes this new gaming device a must have for any avid gamer. Thinking back to the old days of gaming either on a regular PC or one of those early motion gaming consoles gives us that warm fuzzy feeling inside, but no matter how old school we are, the new redefined experience with the GameFloor3 was something we will remember for a long time. 

Cloud computing

The Cloud is already hovering over us. In the simplest of words, cloud computing means shifting some part of your computing devices to a server. This is nothing new, as internet drives have existed for about a decade now. However, the cloud will evolve to take up a greater fraction of the duties – till the only thing on your side of the channel will be a display, and your input devices. What is most important is



that the device at your end could be anything from a laptop to a tiny phone, and it will still compute in seconds! One case of cloud computing, very similar to what we have envisioned here, is OnLive. It is an on-demand cloud gaming server, where your network speed is the only limit. So given the right bandwidth, you could get Crysis 2 running flawlessly on a Pentium 4! With sophisticated stream compression techniques at work, OnLive plans to invade all your devices with games and other computationally intensive applications like 3D modelling and animation. Final piece in the cloud puzzle: Homomorphic Encryption – One reason why major corporations don't want to adopt the cloud is the lack of security: Although data can be encrypted when sent to the cloud server, it cannot process the data in its encrypted form, and thus will have to decrypt it to process it – a proposition that companies are not very comfortable with. So, what was needed was an encryption standard which would allow the server to manipulate the data in its encrypted form. When the resultant is decrypted on client side, it gives the required result. This Homomorphic Encryption could finally lead to mass adoption of cloud tech.

Motion sensing

Who hasn't heard of Kinect by now. Microsoft's motion sensing attachment for Xbox 360 used capture technology from Israeli firm PrimeSense.

Kinect works by creating a 3D model of the room, and the people inside it, using depth sensing cameras and infrared sensors. This model is then analysed by the software part of Kinect to associate body gestures with commands. A completely natural interface, motion sensing is going to be huge in future. Gaming will probably gain the most from further sophistication of motion sensing. The challenges ahead lie in the form of more accurate recognition of more subtle body movements, and doing so while the user is standing or sitting – a position that Kinect does not permit as of now.

